

LED Industry in 2011 and India

The LED industry made about 100 billion LED chips in 2011 for a market that needs only 89 billion. That 12 per cent 'oversupply ratio' will worsen to 21 per cent in 2012 after LED makers add manufacturing capacity. This would mean further drop in prices of LED bulbs and tubes



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Compared to incandescent bulbs and compact fluorescent lamps (CFLs), light-emitting diodes (LEDs) are characterised by a low operating voltage and current, high reliability and long life. The latest range of ultra-bright LEDs consumes just 11 per cent of the electrical power consumed by incandescent bulbs for the same light output in lumens. That means today's 11W LED light can give the same light output as a 100W conventional filament bulb.

Thus the global power demand will reduce by more than half if all conventional lights are replaced with LEDs. This will eliminate the need to use dangerous and controversial nuclear energy or polluting fossil fuel power, and also reduce dependence on other countries for energy security.

Under the shadow of the international energy crisis and gradual improvement of environment protection requirements, semiconductor LED lighting is recognised as the only and most important way of energy saving and environment protection in the world. It lasts long, allows microminaturisation, and is energy-efficient, safe, environment-friendly and available in abundant colours.

LED light sources are very reliable since a single chip converts electrical energy into light energy. As this conversion efficiency is the highest, the heat loss is small and well managed

to protect the chip. The life of an LED is therefore practically unlimited at 50,000 hours. Against this, CFLs have as many as 40 or more parts and rarely complete the claimed lifespan of 5000 hours. Cheap CFLs from China are indeed a waste of money.

With gradual enhancement of luminous efficiency and application technologies, the application area of LEDs has increased from just indicator lights to displays, landscape lighting, backlight, automotive lighting and traffic lights. LED applications are now in different development phases.

Global LED industry

Globally, the LED industry is growing rapidly. According to a report on the global and Chinese LED industry published by China Research and Intelligence in 2009, "The entry barriers in the light-emitting diode industry are being cut down gradually. The upstream industry includes monocrystalline chip and the epitaxial wafer, while the middle-stream industry is mainly engaged in chip processing, and the downstream industry handles packaging, testing and application. The upstream and middle-stream industries—the most competitive and risky fields in the world market—have more technical ingredients and require more investments. In the LED industry, epitaxial wafer and chip account for about 70 per cent of profits, packaging 10 to

Largest Manufacturers in The World

Company	Ranking
Nichia Corporation	World's largest provider of high-brightness LEDs
Osram Opto	World's second largest and Europe's largest manufacturer of high-brightness LEDs
Toyoda Gosei Co.	World's fourth largest and Japan's second largest manufacturer of LEDs